Who Counts as a Neighbor?

From empty regions to AI, topology, and stochastic geometry

Héctor Maravillo (UAM, Mexico) **Diego Villarreal de la Cerda** (T-Systems)
Heriberto Espino-Montelongo (Universidad de las Américas Puebla)

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A cloud of points has no edges until someone defines what *neighbor* means. That definition determines which information can flow through a graph, which gaps count as meaningful, and which geometric structures survive noise. It may also determine the output of a graph neural network or the topology inferred from a dataset. **ProximityGraphs** is an open-source Python toolkit that turns this modeling decision into an experiment: construct multiple geometric graphs on the same point set, compare their structure, and examine what changes when the neighborhood rule changes.

From locations to structure

Voronoi diagrams divide space by the nearest site; their Delaunay dual connects sites with adjacent cells. The same construction can organize emergency-response territories, facility allocation, spatial interpolation, and adaptive meshes for partial differential equations. John Snow's 1854 cholera investigation provides a historical setting: the HistData digitization supplies pump positions and individual death-location marks. We construct a **modern Euclidean overlay** on those coordinates, using pumps as sites and deaths only as contextual observations. Snow did not compute this diagram; straight-line proximity is not measured travel or evidence of causation.

Figure 1 · John Snow's pumps: Voronoi and Delaunay

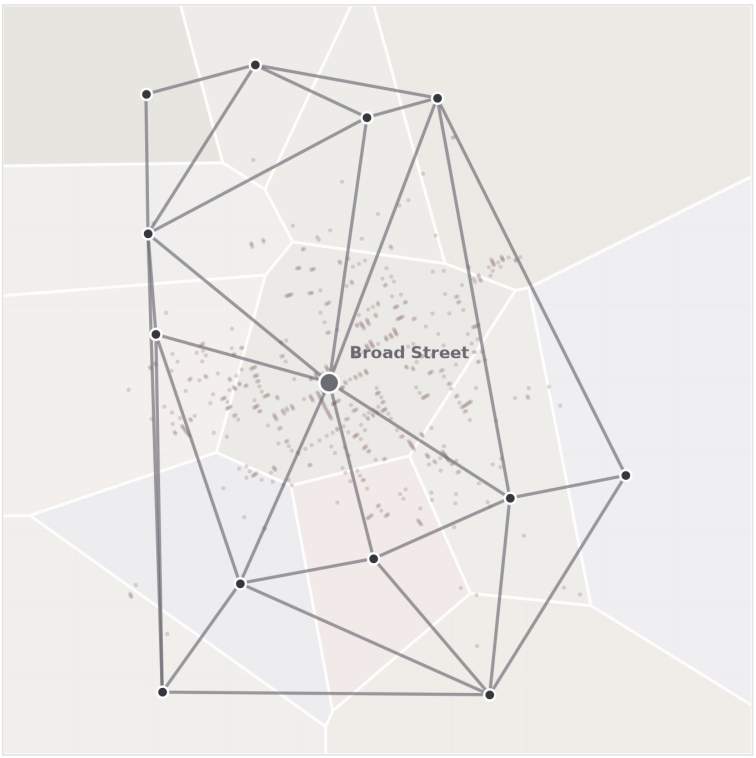


Figure 1: *Historical observations, modern geometric partition.* Thirteen digitized Soho water pumps—including Broad Street—define Voronoi cells and Delaunay edges; 578 cholera death-location marks remain a separate, fixed layer. Readers can reveal pump labels, nearest-site bisectors, and empty Delaunay circumcircles. Moving a pump produces a hypothetical configuration. Data: HistData's digitization of John Snow's 1854 map; visualization: the authors using ProximityGraphs Explore the interactive figure.

The same tension governs road networks. Aldous and Shun (2010) quantify the conflict between construction length and travel efficiency, while Maravillo Gómez and collaborators (2023) show how β-skeletons recover idealized cyclic street tilings

and approximate real urban patterns. Sparse adjacency thus encodes more than appearance: it determines whether useful routes require costly detours.

Away from geography, images, gene-expression profiles, astronomical spectra, and learned embeddings are points in high-dimensional feature spaces. Costa and Hero (2004) estimate intrinsic dimension and entropy from geodesic minimum-spanning-tree lengths; more directly, Dyballa and Zucker (2023) initialize adaptive manifold neighborhoods with a **Gabriel graph**, then sparsify it to improve geodesic computation and dimension estimation. Choosing neighbors is already part of the learning algorithm, not merely a preliminary visualization.

The general mathematical setting is $\mathbb{R}^d$**, for any** $d \geq 1$: points can be geographic coordinates, 3D positions, gene-expression measurements, or learned embeddings. The current **ProximityGraphs implementation concentrates on planar algorithms**; although its point-set container accommodates other coordinate dimensions, not every graph constructor is yet dimension-general.

Shape, holes, and scale

A convex hull covers a point cloud but hides indentations. Alpha-shapes and circular-arc **alpha-hulls** recover scale-dependent envelopes: a radius threshold can expose bays, disconnected pieces, and holes. Here a Delaunay triangle survives when its circumradius is at most $R = 1/|\alpha|$. The resulting straight alpha-shape boundary contrasts with the circular arcs of the related alpha-hull. Such summaries inform shape reconstruction, microscopy, porous-material analysis, and spatial delineation—but are not automatically confidence regions or physical surfaces.

Figure 2 · Image-traced serif Pg · alpha-shape counters

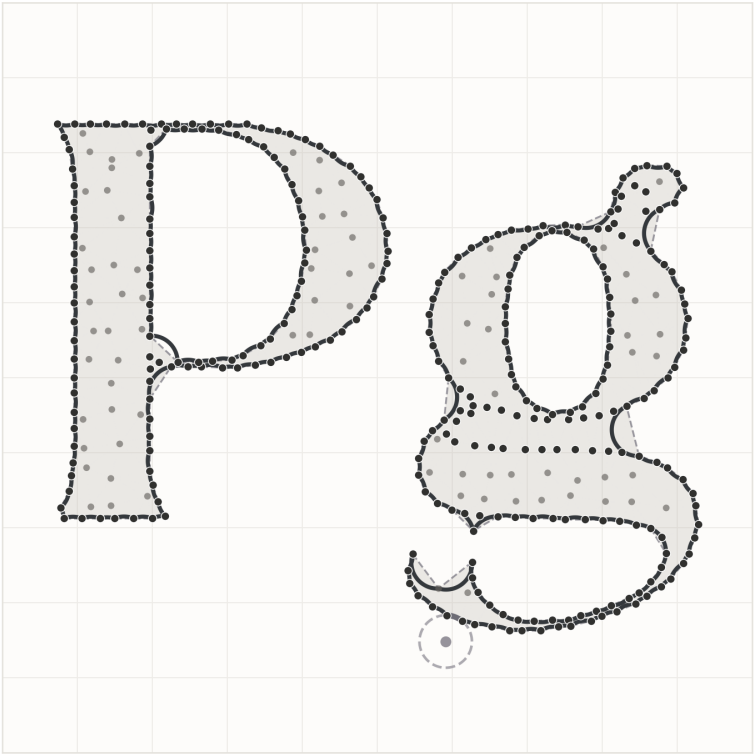


Figure 2: *Letters, holes, and circles.* A synthetic **Pg** cloud contains 600 sites by default: 298 fixed contour sites and 302 interior sites (including 220 seeded-random samples). At $\alpha \approx -28.57$, two separate letter components retain two genuine counters. Readers can select **0–600 interior points** with a density slider, reshuffle random samples, change the alpha scale, and inspect the straight alpha-shape boundary or the curved alpha-hull and its supporting circles. The 53-point smiley remains available as an alternative. Point sampling and graphic: the authors Explore the interactive figure.

Topology makes the scale question more consequential. **Alpha complexes**, related to but distinct from the displayed alpha-hulls, form filtrations in which persistent homology tracks connected components, loops, and higher-dimensional voids. Niyogi, Smale, and Weinberger (2008) establish conditions under which homology of an unknown manifold can be recovered reliably from random samples. Such ideas connect molecular cavities, porous materials, brain imaging, and the topology of learned representations. Graph geodesics similarly underpin nonlinear dimensionality reduction, where one misplaced edge

can create a false shortcut between manifold folds. **The inferred structure is conditional on the neighborhood model.** ProximityGraphs provides graph constructions for investigating that choice, not a built-in persistent-homology engine.

Empty regions as computational decisions

In a **beta-skeleton**, two points are joined when a prescribed region between them contains no competing point. The empty lune for $\beta = 2$ yields the relative neighborhood graph; at $\beta = 1$, the Gabriel graph arises. For planar point sets in general position, a familiar hierarchy is

$$\mathrm{MST} \subseteq \mathrm{RNG} \subseteq \mathrm{GG} \subseteq \mathrm{Delaunay}.$$

The inclusions expose different degrees of geometric sparsity, which matters for sensor-network candidates, decentralized communication, geometric routing, spatial anomaly detection, image morphology, and the skeleton of an observed shape. They do **not** guarantee that a wireless edge is physically viable or that a graph is an optimal routing network.

Figure 3 · Relay corridor: beta-skeleton links

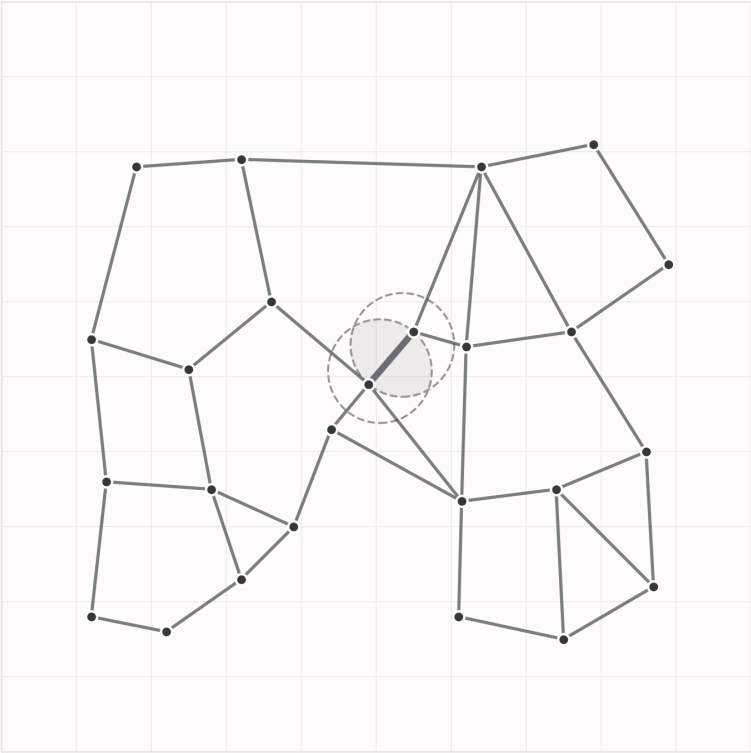


Figure 3: *An edge justified—or rejected—by emptiness.* Twenty-five synthetic sensor sites form two clusters joined by three relays. Select any candidate pair to display its two disks and their intersecting lune: an accepted edge has no blocker, whereas a rejected pair is dashed red and its blocking site is highlighted. Move or add a site to change the test. Synthetic layout and graphic: the authors using ProximityGraphs Explore the interactive figure.

This selective adjacency reaches AI systems through **inductive bias**: in a graph neural network, the edges define which objects exchange messages. In transformer-based search, neighborhood structures support approximate retrieval of relevant text, images, and other embeddings; Iff and collaborators (2026) benchmark filtered nearest-neighbor methods on embeddings from more than 2.7 million research papers, a setting relevant to retrieval-augmented generation and recommendation. The underlying question is shared—**which neighbors preserve useful information without making every pair adjacent?**—but ProximityGraphs is a geometric-graph toolkit, not a transformer index or a vector database.

One-line stochastic geometry

What happens when the sites themselves are random? Devroye (1988) studied empty-region graphs in $\mathbb{R}^d$ and showed that remarkably simple asymptotic degree laws emerge across broad sampling densities. For a stationary Poisson process of

intensity ρ, suppose the exclusion region of a pair at separation r has volume $a\,r^d$. Writing v_d for the unit-ball volume gives

$$\Pr(\text{edge} \mid r) = e^{-\rho a r^d}, \qquad \mathbb{E}[\deg] = \frac{v_d}{a}.$$

Thus **shape enters through a unit-region volume, and the mean degree does not depend on point intensity** under this homogeneous model. In two dimensions, the Gabriel graph has mean degree **4**; the relative neighborhood graph has mean degree approximately **2.558**. Penrose and Yukich (2013) show that many local geometric statistics of random samples on a smooth manifold have limiting laws governed by Poisson processes on its intrinsic-dimensional tangent planes. This connects ambient coordinates, hidden dimension, and statistical uncertainty, although curvature corrections beyond the tangent limit remain a separate mathematical problem.

A geometry laboratory in Python

ProximityGraphs combines point generators (including random spatial processes), graph constructors, visualization, graph summaries, and repeated-experiment helpers. For instance, `pg.GG(points)`, `pg.RNG(points)`, and `pg.DelaunayG(points)` let researchers compare adjacency on unchanged coordinates; the resulting graphs can inform, rather than replace, downstream learning or topological analysis. Its source is MIT-licensed, and optional GIS utilities connect the workflow with geospatial data.

The interactive companion turns these illustrations into inspectable computations: move, add, or delete points; expose bisectors, circumcircles, retained simplices, and rejected edges; or compare graph families on identical sensor coordinates. Unedited geometry is exported from the Python library; edited configurations are recomputed independently in the browser. The point is not that one graph always wins, but that **neighbor definitions become assumptions we can see, modify, and test**.

Try the library: Source and examples · PyPI · `pip install proximitygraphs`. The Python release is currently a pre-release; selected algorithms and models are planar.

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About the authors

Héctor Maravillo works on computational geometry and proximity graphs at Universidad Autónoma Metropolitana (UAM), Mexico.

Diego Villarreal de la Cerda works at T-Systems and contributes to the computational development of ProximityGraphs.

Heriberto Espino-Montelongo develops mathematical research in geometric probability, stochastic geometry, and spatial networks at Universidad de las Américas Puebla.